

PROJECT REPORT

EE200

Signals, Systems and Networks

Prepared by

Yagnik Parmar

Practical Supervisor

Prof. Tushar Sandhan

yagnikp24@iitk.ac.in
E-mail Address

241191
Roll Number

EE
Branch

Q1. FREQUENCY MIXER: ‘BEAUTY AND THE BLUR’

1. Abstract:

“This project explores the frequency characteristics of grayscale images using 2D Fourier Transform. Using Python and OpenCV, we analyze and visualize both the original and a 90° rotated version of an image. Magnitude and dB spectra in centered and non-centered forms are plotted. The results show how image content and its frequency components are intrinsically connected. The project enhances understanding of hybrid image construction and image transformation analysis.”

“This practical explores how different spatial frequency components in an image contribute to human perception. Using Fourier Transform techniques, we analyze and manipulate the frequency spectrum of grayscale images. A frequency mixer system is developed in Python to combine the low-frequency structure of one image with the high-frequency details of another. This technique creates a hybrid image that changes in appearance depending on the viewing distance—illustrating how our visual system processes different spatial scales. The experiment involves spectrum computation, visualization, frequency shifting, rotation analysis, and hybrid image synthesis.”

2. Observation:

- To understand and visualize how 2D Discrete Fourier Transform (DFT) reveals image frequency components.
- To rotate and compare the spectral effects of image transformations.
- To create a hybrid image by combining the fine details (high frequency) of one image with the structure (low frequency) of another.
- To plot the transfer functions that represent the image fusion process.

3. Theoretical Background:

3.1 2D Discrete Fourier Transform Equation

For a 2D discrete image $I(x, y)$ of size $M \times N$, its 2D DFT is given by:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x, y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

The inverse DFT is:

$$I(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \cdot e^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

3.2 FFT, Magnitude and dB Spectrum

We compute FFT using `np.fft.fft2()`. The magnitude spectrum is:

$$|F(u, v)| = \sqrt{\text{Re}(F)^2 + \text{Im}(F)^2}$$

The dB spectrum is:

$$20 \times \log_{10}(|F(u, v)| + \varepsilon)$$

Where ε is a small constant to avoid $\log(0)$.

4. Methodology:

4.1 Frequency Analysis of a Single Image:

- Downloaded grayscale image from Google Drive.
- Applied 2D FFT to compute frequency spectra.
- Plotted:
 - i) Non-centered magnitude and dB spectra.
 - ii) Centered spectra using `np.fft.fftshift()`.
- Rotated image 90° anti-clockwise and repeated FFT analysis.
- Compared rotated image's frequency spectra with the original.

4.2 Hybrid Image Construction (Frequency Mixer):

- Two images were resized to a standard shape (512×512).
- Low-frequency part: Gaussian-blurred version of Image A.
- High-frequency part: Amplified subtraction of blurred version from Image B.
- Added the components to create the hybrid image.
- Plotted 2D transfer functions:
 - i) Low-pass (Gaussian)
 - ii) High-pass (1 - Gaussian)
 - iii) Combined transfer function.

5. Code Summary:

Two main scripts were written:

A. Frequency Spectrum Analysis Script:

- Download image → FFT → Plot spectra
- Rotate image → Re-compute FFT → Compare

B. Hybrid Image Creation Script:

- Extract low and high-frequency components
- Combine them
- Normalize and save final hybrid image

6. Results & Observations:

6.1 Spectral Centering:

- By default, FFT output has low-frequency components in the corners.
- Using `np.fft.fftshift()`, the spectrum is centered with low frequencies at the center.

6.2 Rotation Observation:

- Rotating an image by 90° causes its frequency spectrum to rotate by the same angle.
- The spectral energy distribution remains constant, verifying the rotational property of the 2D FFT.

6.3 Hybrid Image:

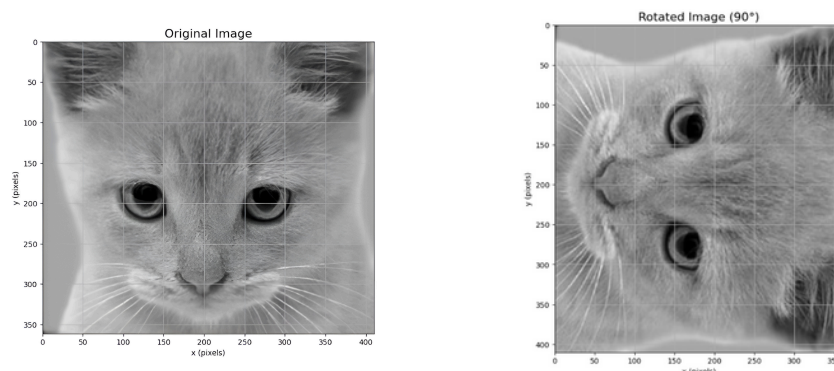
- From a close distance, the viewer sees the high-frequency face (details).
- From far, the low-frequency face dominates perception.
- This confirms how spatial frequencies influence visual recognition.

6.4 Transfer Functions:

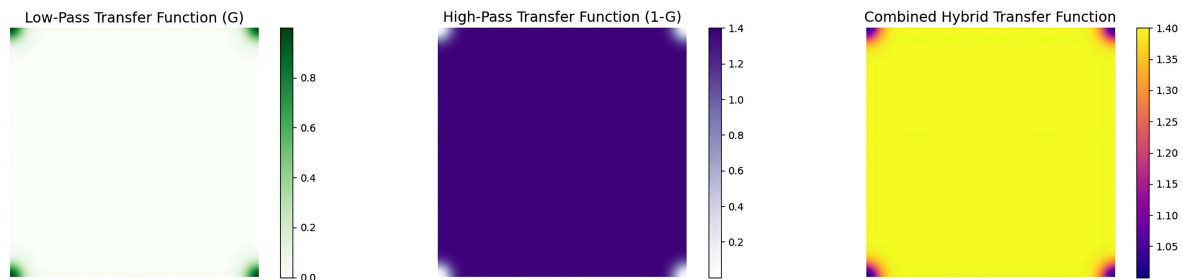
- Gaussian filter effectively models low-pass behavior.
- High-pass component ($1 - \text{Gaussian}$) captures edges and details.
- Combined filter shows the fused effect contributing to both structure and detail.

7. Visual Outputs:

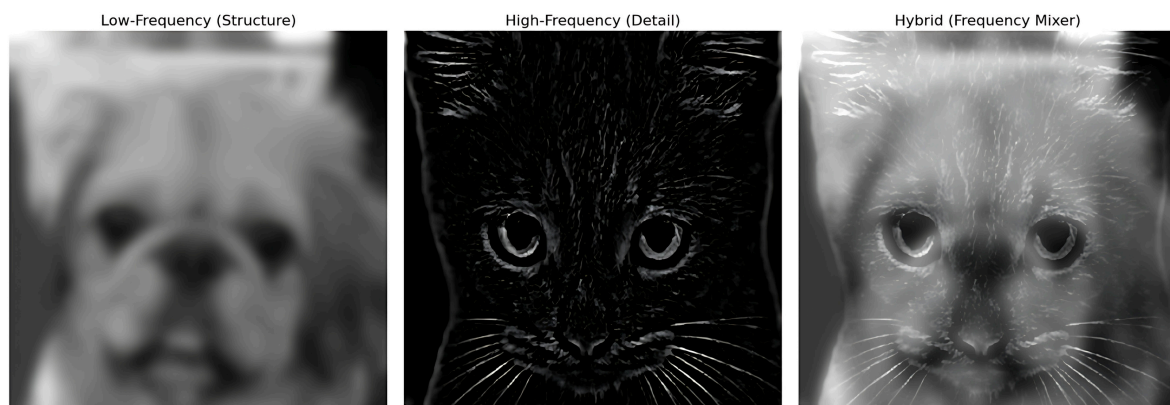
- Original and rotated images



- Magnitude and dB spectra (centered and non-centered) {Can be seen in the python note}
- Gaussian transfer functions (3D filter plots)



- Final hybrid image (side-by-side comparison)



8. Conclusion:

The frequency mixer system developed in this practical demonstrates the utility of the 2D Fourier Transform in visual signal analysis. The experiment confirms that high and low-frequency image components contribute differently to human perception. Hybrid image creation allows us to control and manipulate these frequencies to craft images that change appearance based on viewing scale—a powerful demonstration of signal decomposition.

The End